

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 2

Case Study: Are Commodity Prices and the BRL/USD Integrated? Testing and Modelling Long-Run Trends

Application area: *Finance / Macroeconomics — Commodity Markets*

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FRED (<https://fred.stlouisfed.org>): DCOILBRETEU (Brent crude, daily), GOLDAMGBD228NLBM (gold), DEXBZUS (BRL/USD). All available from 1990s. Use the `fredr` R package or direct CSV download.

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1. Background and Objectives

Brazil is a major commodity exporter (oil, iron ore, soybeans). The BRL/USD exchange rate co-moves with commodity prices over the long run. This tutorial tests whether commodity prices are $I(1)$, identifies the correct trend treatment, and warns against spurious correlation.

Learning objectives: (i) Apply a battery of unit root tests; (ii) identify and treat structural breaks (Perron, 1989); (iii) distinguish trend-stationary from difference-stationary series; (iv) detect and avoid spurious regression.

2. Data Preparation

```
library(fredr); library(urca); library(tseries)
fredr_set_key("YOUR_API_KEY") # free at fred.stlouisfed.org

oil <- fredr("DCOILBRENTU", frequency="m", units="log")
gold <- fredr("GOLDAMGBD228NLBM", frequency="m", units="log")
brl <- fredr("DEXBZUS", frequency="m", units="log")
```

3. Exercises

3.1. Visual inspection and transformation

Plot log Brent crude, log gold, and log BRL/USD on the same figure (monthly, 1995–2024). Identify visually: (a) any apparent stochastic or deterministic trends; (b) any major structural breaks (e.g., 2008 financial crisis, 2020 Covid shock, 2014 commodity crash).

3.2. Unit root tests

For each of the three series:

- Run the ADF test with (i) no constant, (ii) constant only, (iii) constant and trend. Select lag length by BIC.
- Run the KPSS test (level and trend versions).
- Report a 3×3 summary table. What is your conclusion about the integration order of each series?

3.3. Spurious regression detection

Regress log BRL/USD on log Brent crude (levels). Report R^2 and the Durbin–Watson statistic. Apply the ADF test to the residuals using MacKinnon (1991) critical values. Is this a spurious or a genuine long-run relationship?

If residuals are $I(1)$ (ADF fails to reject), the regression is spurious despite a high R^2 . If residuals are $I(0)$ (ADF rejects), the series are cointegrated — a genuine long-run equilibrium.

3.4. Structural breaks and unit roots

Apply the Zivot–Andrews test to log Brent crude allowing for a break in intercept and trend (`ur.za` in R). At what date does the test locate the break? How does this change the unit root conclusion relative to the standard ADF result?

3.5. Trend-stationarity vs. difference-stationarity

For log gold: (a) detrend by regressing on a linear time trend; (b) take first differences. Plot both residuals. Which treatment produces an $I(0)$ series according to ADF + KPSS? What are the implications for long-run forecasting?

4. Core References

Dickey, D.A. & Fuller, W.A. (1979). *JASA*, 74, 427–431. Perron, P. (1989). *Econometrica*, 57, 1361–1401. Nelson, C.R. & Plosser, C.I. (1982). *JME*, 10, 139–162.
